

Bias Incidents in Montgomery County

April 2026

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MCPD Bias Incidents

Target Variables

- Roughly 2000 Current Cases
- 2016 - Now

Key Variables:

- Bias_Code
- District
- Bias
- # of suspects/# of victims
- Status



Census ACS 5 2016 - 2024

Demographics

- Demographics of Montgomery County

API Ingestion

Data Extraction

- Method of collecting Data
- Keeps EDA, analysis, and modeling completely up to date



Tools Used



Google Colab

Main Environment



Python

Programming Language



Github

Product



Data Montgomery

Database

Wrangling

MCPD Bias Incidents:

- Fix dates
- Add binary for resolved vs unresolved
- Add binary for below or above 18
- Add binary for religious or racial

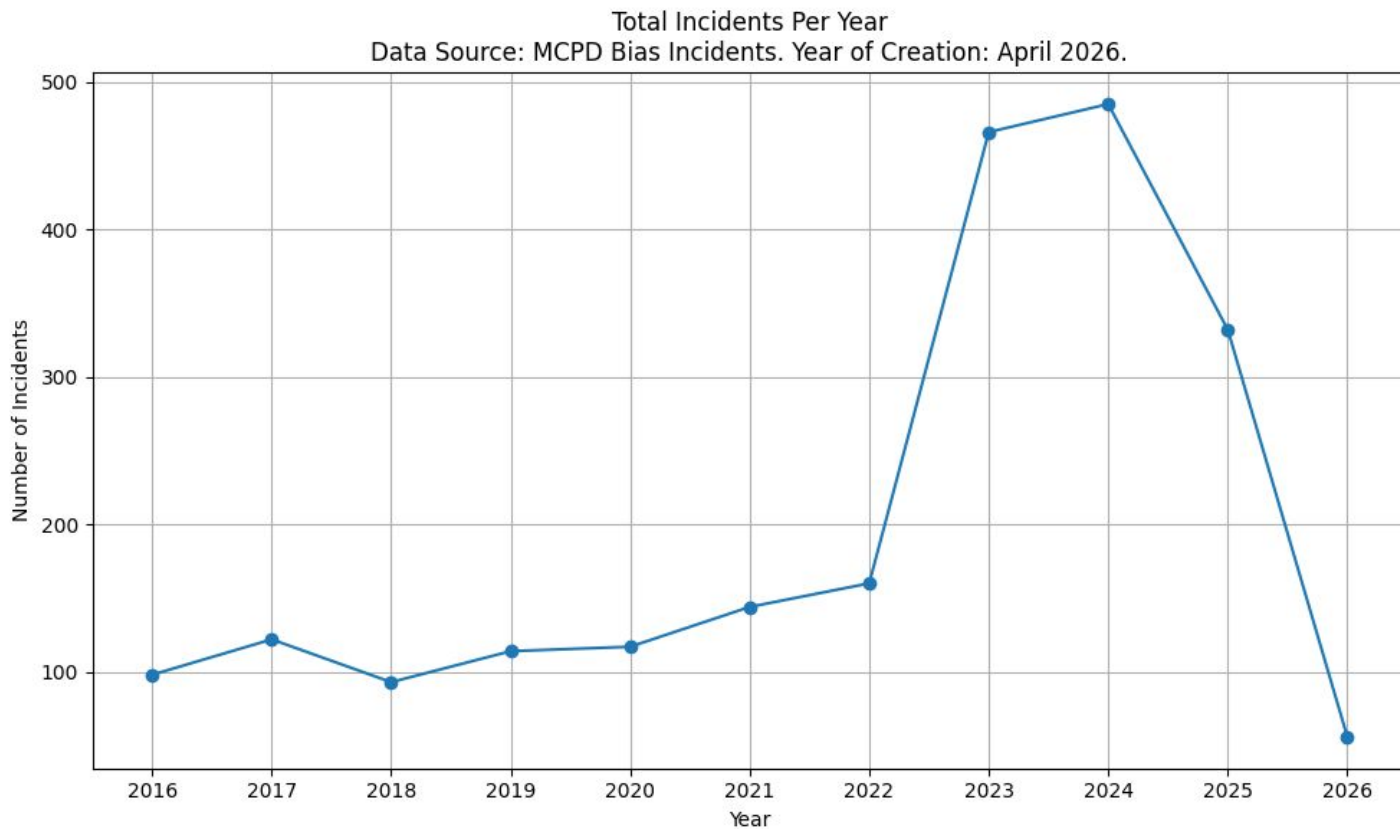
ACS 5:

- Convert districts to cities
- Merge wanted variables to MCPD Bias Incidents

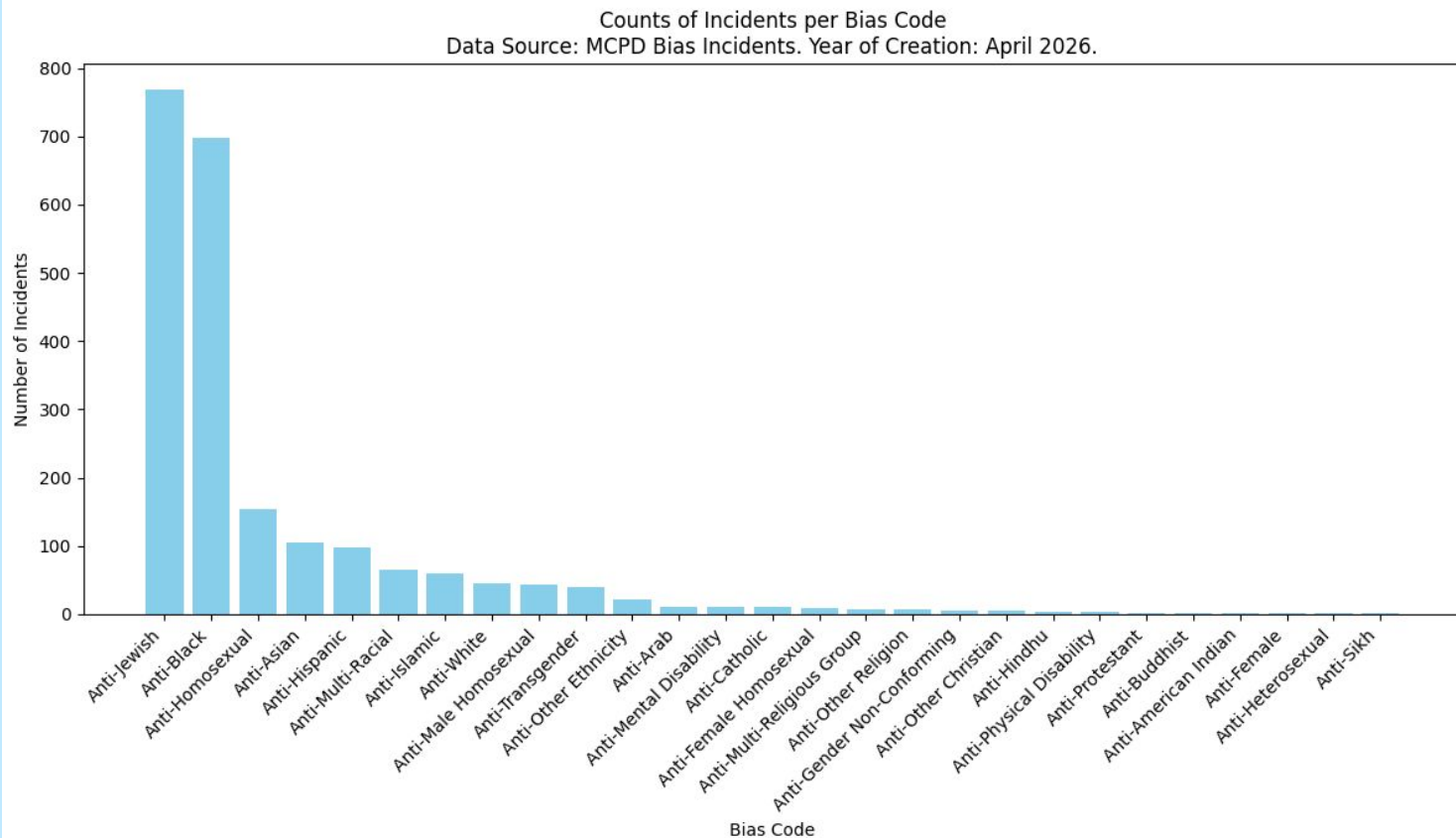
Victim Types Table

victim_type	Count	Percentage
Individual(s)	1186	54.3289
School/College	553	25.3321
Society	164	7.5126
Religious Organization	129	5.9093
Business/Financial Institution	91	4.16858
Government	27	1.23683
Other	19	0.870362

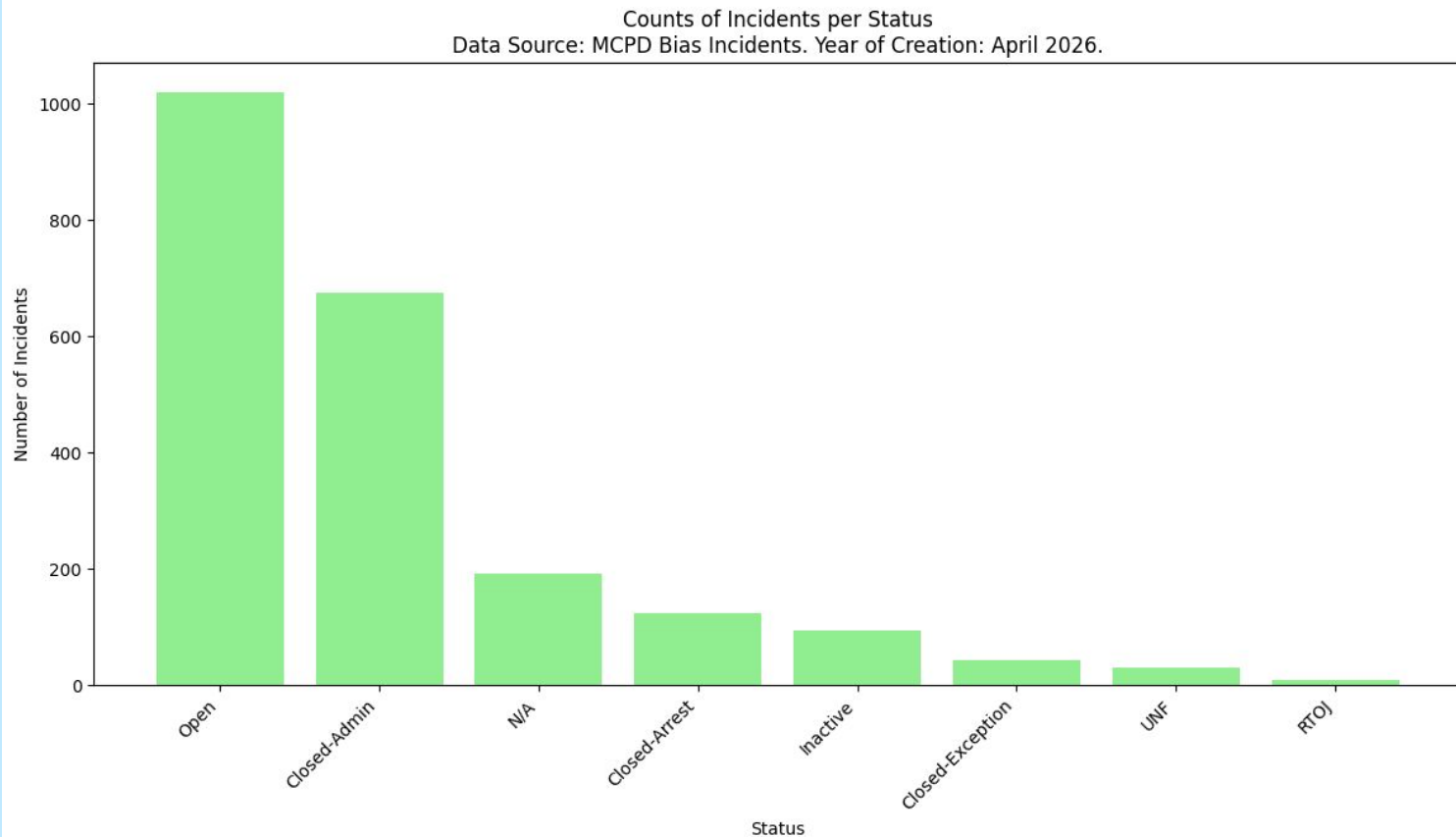
EDA Graphs



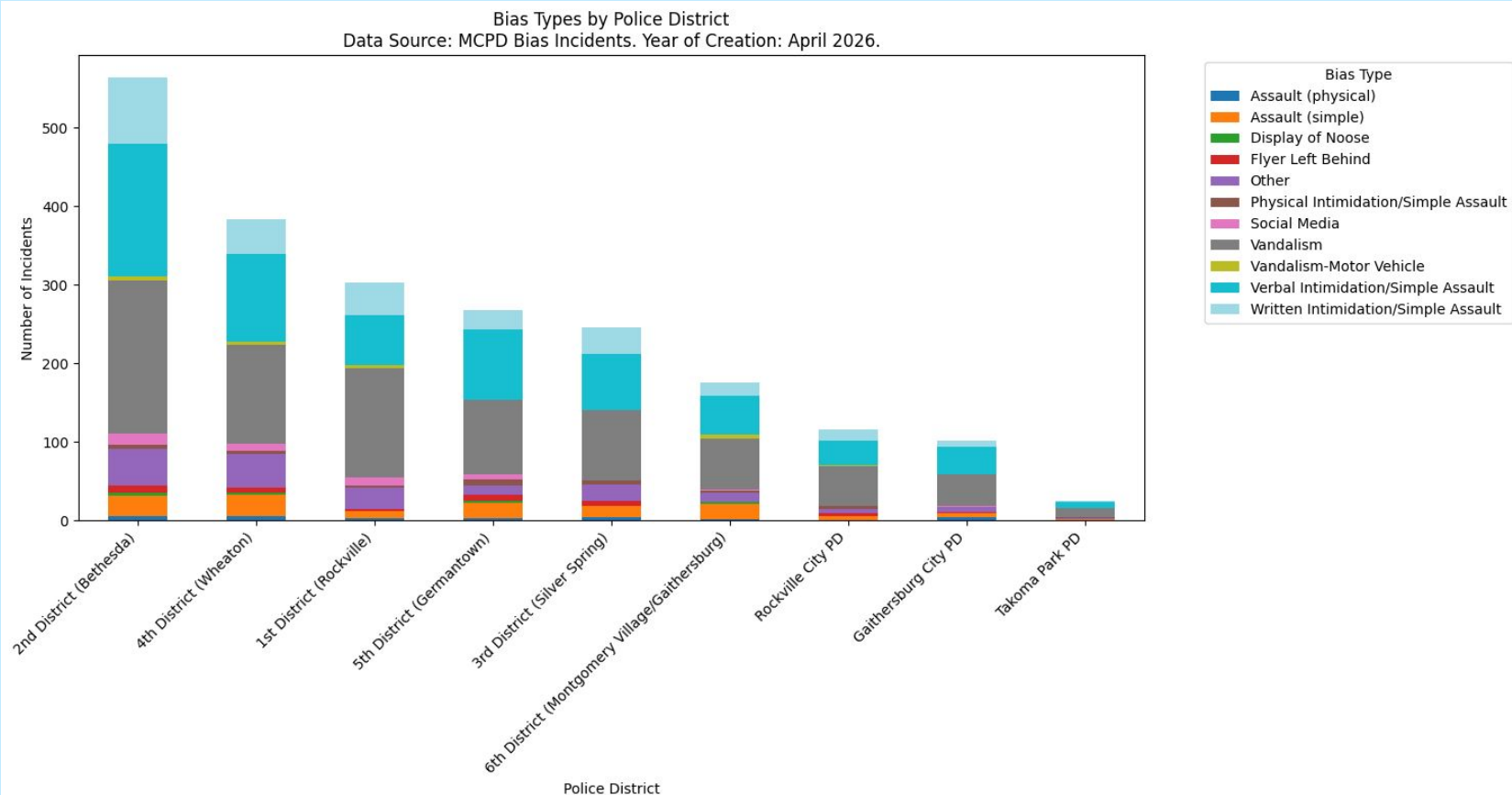
EDA Graphs



EDA Graphs



EDA Graphs



Resolution Rate: Racial vs Religious

Z-Test

Z statistic = -2.07
P-Value = 0.03

Resolution Rates of Religions

Chi-Square

P-Value = 0.22

Resolution Rates of Races/Ethnicities

Chi-Square

P-Value = 0.0002

Representation of under 18 Suspects

Z-Test

Z statistic = -10.67
P-Value = 0.00000 >

Correlation: Population vs Incidents

Pearson s
Correlation

P-Value = 0.36

Correlation: Median Income vs Incidents

Pearson s
Correlation

P-Value = 0.0518

Difference in Means Annual Bias Incidents by City

ANOVA

F statistic = -2.01

P-Value = 0.09

Correlation: Resolution Rate vs Racial Population Proportion

Pearson's
Correlation

Bias Type	ACS Population Category	Pearson Correlation	P-value
Anti-Black	Black or African American Alone Proportion	0.183954	0.183010
Anti-White	White Alone Proportion	-0.100013	0.649800
Anti-Asian	Asian Alone Proportion	0.116055	0.541386
Anti-Hispanic	Hispanic or Latino Proportion	-0.186082	0.316210

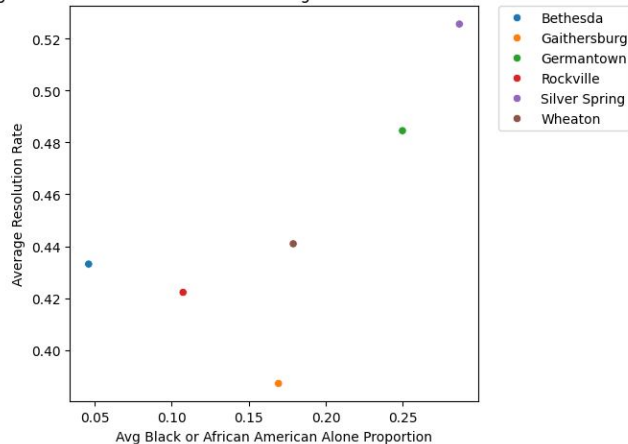
Introduction

Wrangling/EDA

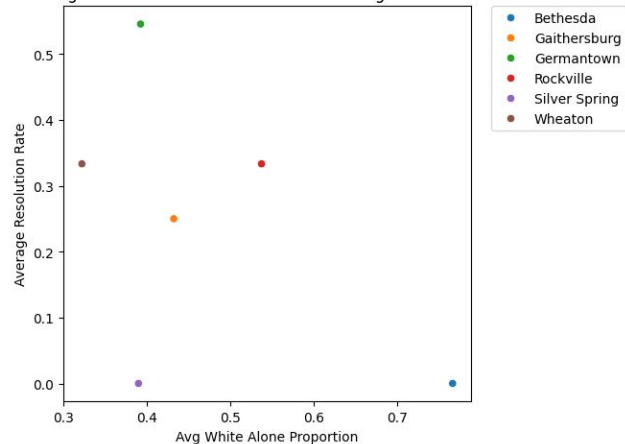
Analysis

Models

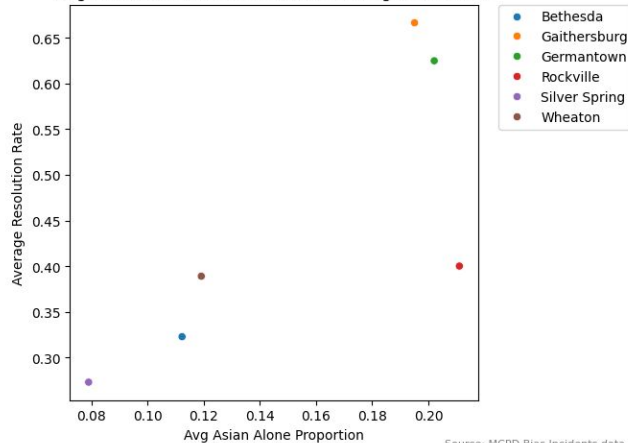
Avg Resolution Rate for Anti-Black vs. Avg Black or African American Alone



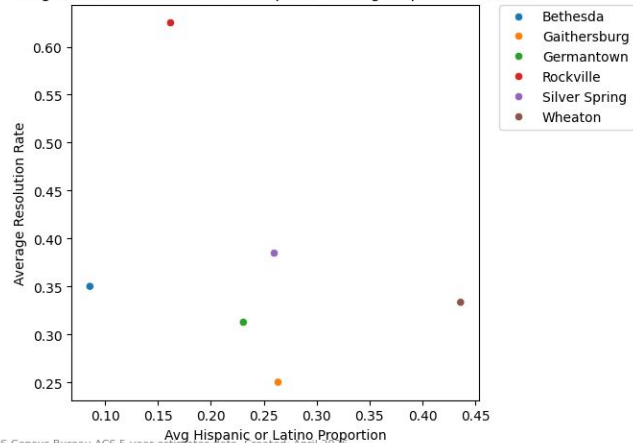
Avg Resolution Rate for Anti-White vs. Avg White Alone



Avg Resolution Rate for Anti-Asian vs. Avg Asian Alone



Avg Resolution Rate for Anti-Hispanic vs. Avg Hispanic or Latino



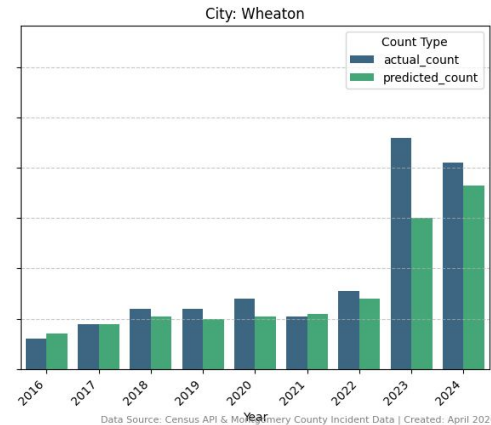
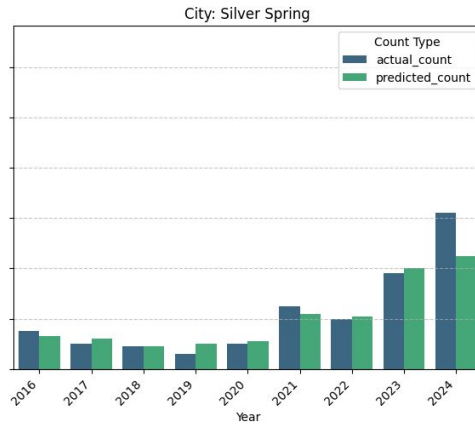
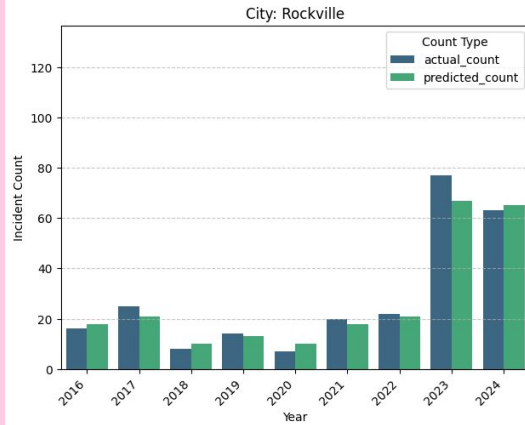
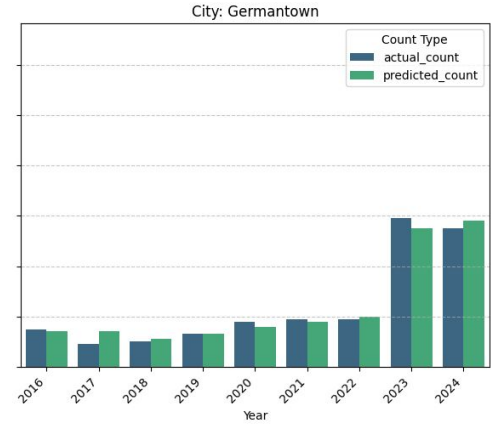
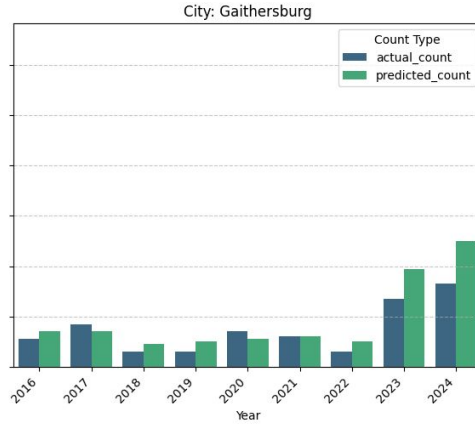
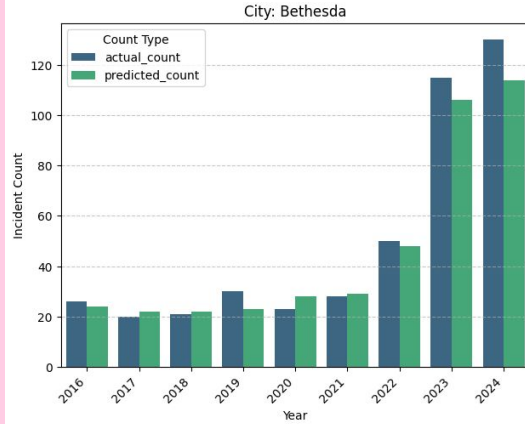
Random Forest Regression

Mean Squared Error: 160.57

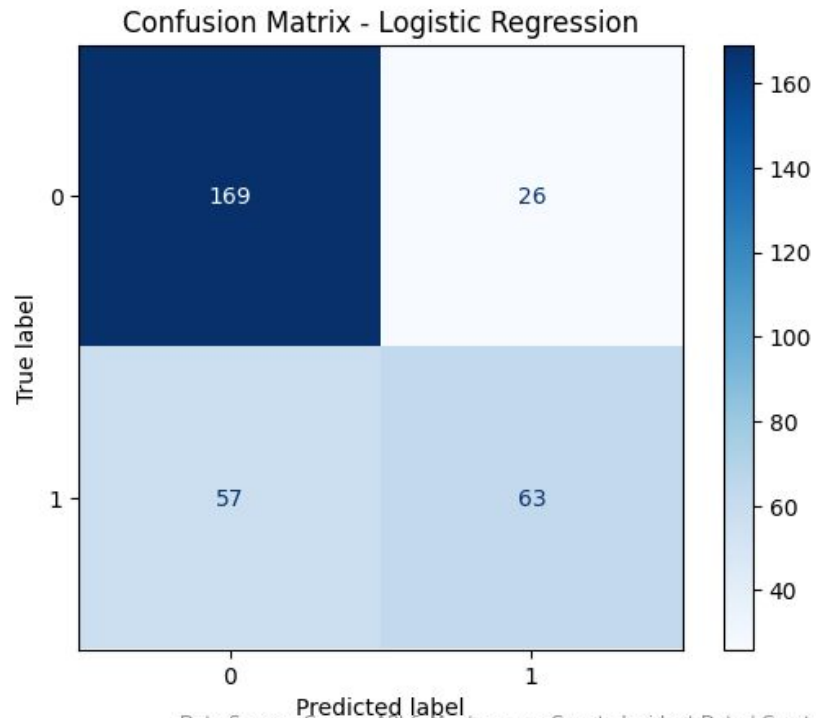
R-squared: 0.72

Feature		Importance
10	incident_year	0.537828
1	Median Household Income	0.271627
0	Total Population	0.040007
4	American Indian and Alaska Native Alone	0.031456
3	Black or African American Alone	0.027824
7	Some Other Race Alone	0.023089
9	Poverty Status (Below Poverty Level)	0.015201
5	Asian Alone	0.015112
2	White Alone	0.014064
6	Native Hawaiian and Other Pacific Islander Alone	0.013274
8	Hispanic or Latino	0.010519

Actual vs. Predicted Bias Incident Counts (2016-2024)

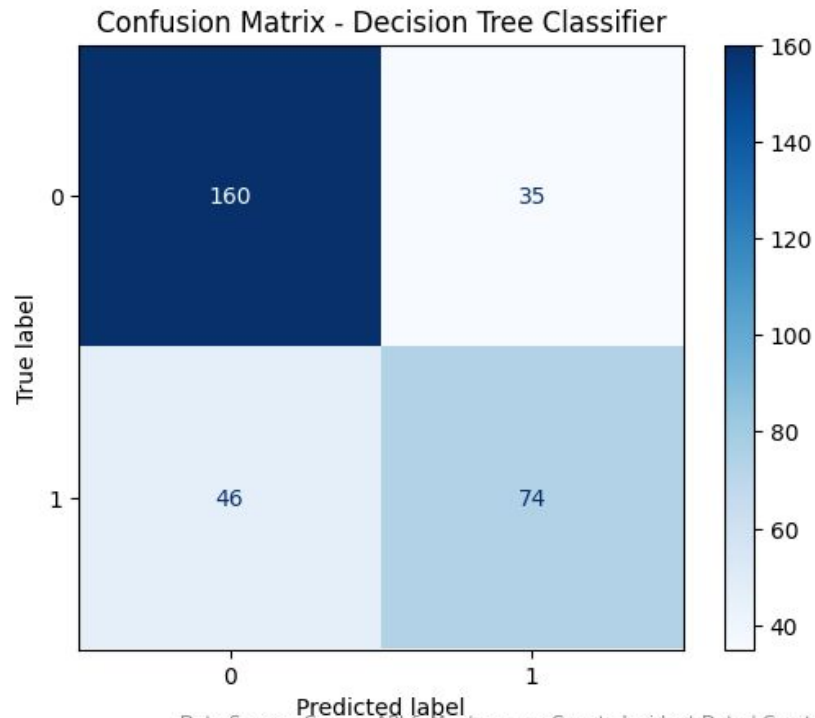


Logistic Regression



Data Source: Census API & Montgomery County Incident Data | Created: April 2026

Decision Tree

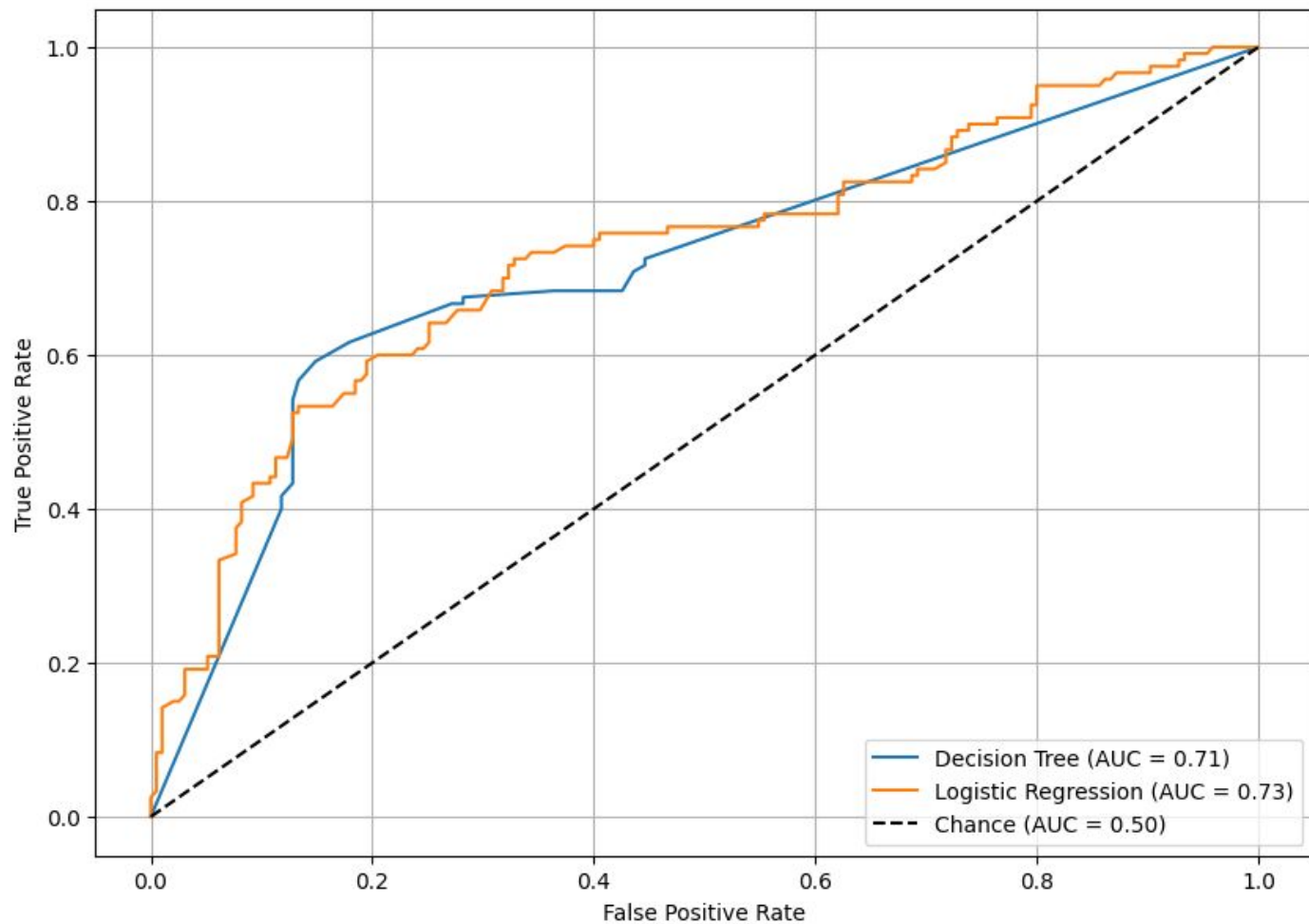


Data Source: Census API & Montgomery County Incident Data | Created: April 2026

Features used:

- Bias_code
- Bias_category
- victim_type
- Under_18_suspect
- Total Population
- Median Household Income
- White Alone
- Black or African American Alone
- American Indian and Alaska Native Alone
- Asian Alone
- Hispanic or Latino
- Poverty Status (Below Poverty Level)

ROC Curve for Case Resolution Prediction



Thank you to...

Professor Saidi,

Professor Alraee

Professor Perine

All of my fellow Data Science Students!

Questions?
